NARAYANA IIT - JEE ACADEMY - INDIA

SR.IIT_*CO-SC(MODEL-A& APEX) GTM-12 Date: 06.01.2025

Time: 3.00Hrs. Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
B	B	A	D	A	C	D	B	A	A
11	12	13	14	15	16	17	18	19	20
A	D	C	C	A	D	B	B	C	D
21	22	23	24	25					
252	9	17	80	3					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	D	B	C	D	D	A	B	B	C
36	37	38	39	40	41	42	43	44	45
B	B	B	B	A	B	A	C	D	A
46	47	48	49	50					
3	59	3	5	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	C	D	A	A	A	B	D	D
61	62	63	64	65	66	67	68	69	70
C	D	A	B	D	B	C	D	A	A
71	72	73	74	75					
4	4	1	1	30					

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HINTS &SOL\
PHYSICS

1. From Gauss's Law

$$\oint \vec{E}.\vec{dA} = \frac{q_m}{\in_0}$$

$$E \cdot 4\pi r^2 = \frac{\int_0^r \left(4\pi r^2 . dr\right)(\alpha r)}{\in_0}$$

$$E \cdot 4\pi r^2 = \frac{4\pi \cdot \alpha \cdot r^4}{4 \in_0}$$

$$E = \left(\frac{\alpha r^2}{4 \in_0}\right).$$

2. Since plane is smooth hence motion is only translational.

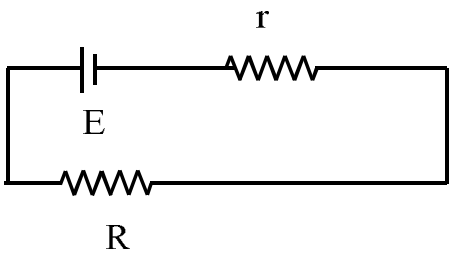
$$a = g\sin\theta = 10 \times \frac{1}{2} = 5 \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$1.6 = 0 + \frac{1}{2} \times 5 \times t^2$$

$$t = \sqrt{\frac{3.2}{5}} = \sqrt{0.64} = 0.8 \text{ sec.}$$

3. The given circuit can be redrawn as follows



Here $r = \frac{r_1 r_2}{r_1 + r_2} = \frac{(1)(1)}{1+1} = \frac{1}{2}\Omega$

and $E = \frac{\Sigma E / r}{\Sigma 1 / r} = \frac{\left(\frac{10}{1}\right) + \left(\frac{10}{1}\right)}{1+1}$

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$= 10V$

Maximum power will be developed across R when

$$R = r = \frac{1}{2}\Omega$$

$$\therefore \quad i_{max} = \frac{E}{R+r} = \frac{10}{\frac{1}{2}+\frac{1}{2}} = 10A$$

$$\therefore \quad P_{max} = i_{max}^2 R = (10)^2 (1/2) = 50W$$

4. Escape velocity $v_e = \sqrt{\frac{2GM}{R}}$

$$v_e = \sqrt{\frac{2G\frac{4}{3}\pi R^3 \times d}{R}}$$

$$\sqrt{2G\frac{4}{3}\pi R^3 \times d} = R\sqrt{\frac{8}{3}\pi G d}$$

where $d =$ mean density of earth

$$\because v_e \propto R\sqrt{d}$$

$$\therefore \frac{v_e}{v_p} = \frac{R_e}{R_p}\sqrt{\frac{d_e}{d_p}}$$

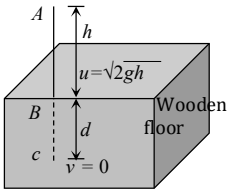
$$= \frac{R_e}{2R_e}\sqrt{\frac{d_e}{d_e}}$$

$$= v_p = 2v_e$$

$$= 2 \times 11 = 22 \text{kms}^{-1}$$

5. $2 \to 1$ in H-atom has same λ as $2 \to 4$ in He-atom.

6. Let the blade stops at depth d into the wood



$$v^2 = u^2 + 2aS$$

$$\Rightarrow 0 = \left(\sqrt{2gh}\right)^2 + 2(g-a)d$$

By solving $a = \left(1+\frac{h}{d}\right)g$

So the resistance offered by the wood $= mg\left(1+\frac{h}{d}\right)$

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7. Area of ellipse = πab

Magnetic dipole moment = $\dfrac{\pi abi}{2}$

Torque $\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB\sin\theta$

$\because \vec{M} \uparrow\downarrow \vec{B}$ $\quad \boxed{\tau = 0}$.

8. Detector will receive maximum light when the image of the source gets formed at the detector itself. For any general time,

Position of source from lens $= -\left[10\text{cm} - (3\text{cm/s} - 1\text{cm/s})t\right] = -\left(10\text{cm} - (2\text{cm/s})t\right)$

Position of detector from lens $= (3\text{cm/s} - 1\text{cm/s})t = (2\text{cm/s})t$

Using the lens formula,

$$\frac{1}{0.9} = +\frac{1}{(10-2t)} + \frac{1}{2t}$$

$$(10-2t)2t = 9$$

$$4t^2 - 20t + 9 = 0$$

$t = 0.5$ sec and $t = 4.5$ sec.

9. As intensity remains same $\Rightarrow$ no of photons remain same

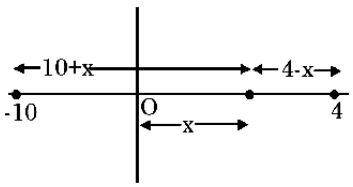
$\Rightarrow$ saturation current will remain same.

Also , as photon energy increases ,

$\Rightarrow$ K.E (max) increases

$\Rightarrow$ stopping potential increases

10.



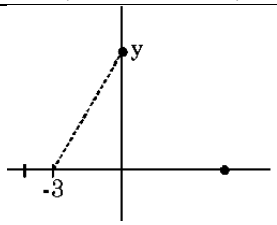
$$t = \frac{10+x}{C} = \frac{4-x}{C}$$

$\therefore -6 = 2x$

$\therefore \quad x = -3$

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$\therefore 7 = \sqrt{3^2 + y^2}$

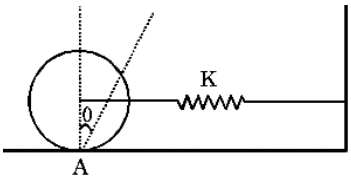
$\therefore y = \sqrt{40}.$

11. In position 2,

$I = \dfrac{E}{R_2}$

$U_{\text{Inductor}} = \text{Heat loss in } R_1 = \dfrac{1}{2} L \left(\dfrac{E}{R_2} \right)^2.$

12.



$\tau_A = I\ \alpha$

$K \times R = \dfrac{3}{2} MR^2\ \alpha$

$KR\ \theta\ R = \dfrac{3}{2} MR^2\ \alpha$

$\alpha = \dfrac{2K}{3M} \theta$

$\therefore \quad \boxed{T = 2\pi \sqrt{\dfrac{3M}{2K}}}.$

13. $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{E_2}{20} = \left(\frac{2T}{T}\right)^4 = 16 \Rightarrow E_2 = 320\ kcal/m^2 min$

14. Dimensionally $F = \dfrac{x}{e}$

$\dfrac{\mathrm{M^1L^1T^{-2}}}{1} = \dfrac{x}{\mathrm{M^1L^{-3}}}$

then $x = \mathrm{M^2L^{-2}T^{-2}}$

and $y = F$

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$y = M^1L^1T^{-2}$.

15. $T - mg = m \times 0$

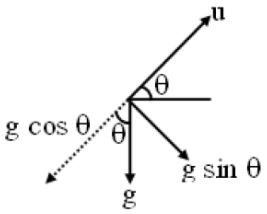
$T = 2000$ N

$W = FS \cos\theta = 2000 \times 40 \times -1$

$W = -8 \times 10^4$ J.

16. R.O.C $= \dfrac{u^2}{g\sin\theta}$

$a_t = g \cos\theta$



so $\dfrac{a_t}{\text{R.O.C.}} = \dfrac{g\cos\theta}{u^2} \times g \sin\theta.$

17.

As $\tau = I\alpha$

$\alpha \propto \frac{1}{I}$ (τ is constant)

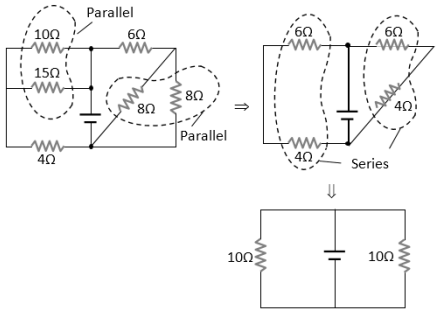
MI of figure (ii) is smaller hence acceleration is greater

18. In case of soap bubble

$W = T \times 2 \times \Delta A$

$= 0.03 \times 2 \times 40 \times 10^{-4} = 2.4 \times 10^{-4}$ J

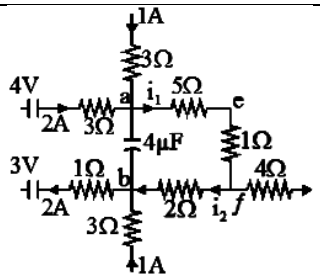
19. Given circuit can be reduced to a simple circuit as shown in figures below



20. When the capacitor plates get fully charged, there will be no current in branch *ab*. Remember capacitance acts as the open circuit since capacitance offers infinite resistance to d.c. The capacitance simply collects the charge.

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Applying Kirchhoff's first law to the junctions a and b, we find $i_1 = 3\text{A}$ and $i_2 = 1$ A
Now applying Kirchhoff's second law to the closed mash $a\ e\ f\ b\ a$, we get

$3 \times 5 + 3 \times 1 + 1 \times 2 = V_a - V_b$

or $V_a - V_b = 20$ V

$$U = \frac{1}{2} C\left(V_a - V_b\right)^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (20)^2 = 8 \times 10^{-4} \text{ J.}$$

21.

$n_A =$ Known frequency $= 256\ Hz, n_B =?$
$x = 4\ bps$, which is decreasing after loading $(i.e.\ x \downarrow)$ also known tuning fork is loaded so $n_A \downarrow$
Hence $n_A \downarrow - n_B = x \downarrow \quad \rightarrow$ correct ...(i)
$n_B - n_A \downarrow = x \downarrow \quad \rightarrow$ Wrong ...(ii)

$$\Rightarrow n_B = n_A - x = 256 - 4 = 252\ Hz$$

22. $7 \times \frac{D\lambda}{d} = \frac{\left(\mu_1 - \mu_2\right)t\ .\ D}{d} \quad t = 9\ \mu\text{m.}$

23. $W_{A \to B} = -22.17$ J

$\Delta U + W = 0$

$\Delta U = -W = 22.17$ J

In second process,

$Q = 9.35 \times 4.19$ J

$Q = W + \Delta U$

'ΔU' will remain same.

$W = Q - \Delta U = 17$ J.

24. $v = \sqrt{\frac{T}{\mu}} = \frac{\omega}{k}$

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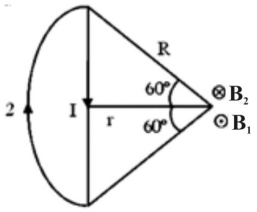
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$$\therefore \quad T = \mu\left(\frac{\omega}{k}\right)^2$$

$$= (0.2)\left(\frac{420}{21}\right)^2$$

$$= 80N$$

25. $r = R\ \cos\ 60^\circ$



$$B_{net} = B_1 - B_2$$

$$= \frac{\mu_0 i}{4\pi r}(\sin\alpha + \sin\beta) - \frac{\mu_0 i}{2r}\left(\frac{\theta}{2\pi}\right)$$

$$= \frac{\mu_0 i\left(\sin 60^\circ + \sin 60^\circ\right)}{4\pi R\cos 60^\circ} - \frac{\mu_0 i}{2R}\left(\frac{2\pi/3}{2\pi}\right)$$

$$= \frac{\mu_0 i\sqrt{3}}{2\pi R} - \frac{\mu_0 i}{6R} = \frac{\mu_0 i}{6\pi R}\left(3\sqrt{3} - \pi\right).$$

CHEMISTRY

26. The two equation are :

$$\frac{4}{3}\mathrm{Al(s)} + \mathrm{O_2(g)} \rightarrow \frac{2}{3}\mathrm{Al_2O_3(s)},\ \Delta_f G^\circ = -\frac{2}{3}\times 827\mathrm{kJ\ mol^{-1}} \quad \ldots(i)$$

$$\frac{4}{3}\mathrm{Cr(s)} + \mathrm{O_2(g)} \rightarrow \frac{2}{3}\mathrm{Cr_2O_3(s)},\ \Delta_f G^\circ = -\frac{2}{3}\times 540\mathrm{kJ\ mol^{-1}} \quad \ldots(ii)$$

Subtracting equation (*ii*) from equation (*i*) we have,

$$\frac{4}{3}\mathrm{Al(s)} + \frac{2}{3}\mathrm{Cr_2O_3(s)} \rightarrow \frac{2}{3}\mathrm{Al_2O_3(s)} + \frac{4}{3}\mathrm{Cr(s)},\ \Delta_r G^\circ$$

$$= -287\mathrm{kJ\ mol^{-1}} \times \frac{2}{3} = -191.$$

27. In $Fe(CO)_5$, Fe is dsp^3 hybridized.

28. $SbCl_5$ act as a Lewis acid.

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29. Reduction of ketone to alcohol by hydrogen nickel followed by acidic dehydration of alcohol to give alkene.

30. More is the non-volatile solute added, lesser is the vapour pressure. After solubility precipitate formation takes place.

31. $pH_1 = pK_A + \log\frac{1/5}{4/5}$

$pH_2 = pK_A + \log\frac{4/5}{1/5}$

$pH_1 - pH_2 = \log\frac{1}{4} - \log 4$

$\boxed{\Delta pH = 2\log 4}$.

32. Conceptual

33. CH_2^+ form after removal of Cl^- is more stabilized.

34. Trans compound + syn addition → gives racemic mixture.

35. Carbocation stability order is 3°> 2°> 1°.

36. Conceptual

37. $W_{solute} = 3$ gms, $W_{solution} = 50$ gm $= 75 - 25$

$W_{OH^-} = \frac{3}{40} + \frac{X}{56}.\ m = 5 = \frac{\left(3/40 + X/56\right)}{47} \times 1000$

$X = 9.$

38. 3° carbocation is more stable.

$2HSO_4^- \longrightarrow H_2S_2O_8 + 2e^-$.

39. $K_p = K_c\left(RT\right)^{\Delta n}$

$\log\left(\frac{K_P}{K_C}\right) = \Delta n \log RT \quad \Rightarrow \Delta n = 2.$

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40. When a solution containing a non volatile solute is heated in a open vessel, the solution boils at a pressure equal to atmospheric pressure which remains constant. Due to increase in concentration of solution boiling point increases.

41. $\Delta H = -1500$ kJ

$6\Delta H_{P-P} + 3\Delta H_{O=O} - 12\Delta H_{P-O}$

$-1500 = 6 \times x + 3(500) - 12(350)$

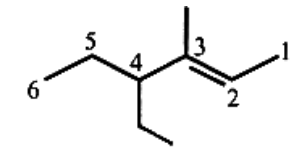
$x = 200.$

42. H_2O has *H*-bonding while H_2Te has higher Vander wall forces than $H_2Se > H_2S$.

43.

List-I		List-I	
(P)	$\psi_{MO} = \psi_A - \psi_B$	(III)	Anti-bonding molecular orbital
(Q)	$\mu = Q \times r$	(I)	Dipole moment
(R)	$\frac{N_b - N_a}{2}$	(IV)	Bond order
(S)	$\psi_{MO} = \psi_A + \psi_B$	(II)	Bonding molecular orbital

44.



4-Ethyl-3-methylhex-2-ene

45. It is a fact that Amylopectin have 1 → 4 α-glycosidic linkage and 1→ 6 α-glycosidic linkage

46. Ozonolysis of double bond to give diketone and then aldol condensation to give a product with 7 member and 5 member ring.

47. $H_2 \longrightarrow 2H^+ + 2e^-$

1 atm 10^{-10}

$E_{H_2/H^+} = 0 - \frac{0.059}{2}\log\frac{(10^{-10})^2}{1}$

$E_{H_2/H^+} = +0.59$ V.

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48. (2 cis and 1 trans)

49. Number of amino groups $= \dfrac{390-180}{42} = 5$

50. For zero order, $t_{1/2}$ decrease with decrease in concentration of reactions.
For first order, $t_{1/2}$ is independent of concentration of reaction.
For second order, $t_{1/2}$ increase with decrease in concentration of reactant.

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{[A_0]_2}{[A_0]_1}\right)^{1-n} \Rightarrow \frac{100}{50} = \left(\frac{0.5}{1.0}\right)^{1-n}$$

$$\Rightarrow 2 = \left(\frac{1}{2}\right)^{1-n} \Rightarrow 2 = 2^{n-1} \Rightarrow n-1 = 1 \Rightarrow n = 2.$$

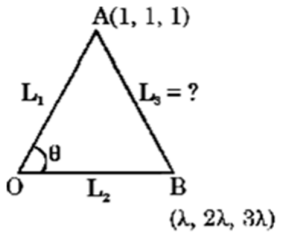
MATHEMATICS

51. First element in 20th group is equal to '20;.
(20, 21, ….. upto 39 terms)

$$\therefore \text{ Sum} = \frac{39}{2}\left(2\times 20 + (39-1)\times 1\right)$$

$$= \frac{39}{2}(40+38) = (39)^2.$$

52.



$\cos\theta = \dfrac{6}{\sqrt{42}} \Rightarrow \sin\theta = \dfrac{\sqrt{6}}{\sqrt{42}}$ (where θ is the angle between L_1&L_2)

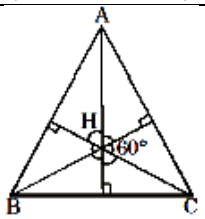
Area $\Delta OAB = \dfrac{1}{2}(OA)(OB)\sin\theta$

$$= \frac{1}{2}\left(\sqrt{3}\right)|\lambda|\left(\sqrt{14}\right)\frac{\sqrt{6}}{\sqrt{42}} = \sqrt{6} \Rightarrow \lambda = \pm 2.$$

53.

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The two altitudes are

$\left(x-\sqrt{3}y+1\right)\left(x+\sqrt{3}y-5\right)=0$

Point of int. of the 2 altitudes is $\left(2,\ \sqrt{3}\right)$

Let slope of 3rd altitude be 'm'

then $\left|\dfrac{m-\dfrac{1}{\sqrt{3}}}{1+\dfrac{m}{\sqrt{3}}}\right|=\sqrt{3}\Rightarrow\dfrac{\sqrt{3}m-1}{\sqrt{3}+m}=\pm\sqrt{3}$

$\sqrt{3}m-1=\pm3\pm\sqrt{3}m$

$m=\infty,\ m=\dfrac{-2}{2\sqrt{3}}=\dfrac{-1}{\sqrt{3}}$

$\therefore$ The third altitude is $x = 2$.

54. $56=\dfrac{50n+60m}{m+n}$

$56n+56m=50n+60m$

$6n=4m$

$\dfrac{n}{m}=\dfrac{2}{3}$

$\therefore 9\ .\ \dfrac{n}{m}=9\times\dfrac{2}{3}=6.$

55. $A^2\,|A|^2-|B|^{3-2}\,B$

$=25A^2-|B|.B$

$=25A^2+|B|A^2$

$=0$

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$$\left[\begin{array}{l}\because\ A^{-1}B^2+AB=0\\ \text{Pre - multiply by 'A'}\\ B^2+A^2B=0\\ B^2=-A^2B\\ \text{Post multiply by } B^{-1}\\ B=-A^2\\ |B|=-|A|^2\\ |B|=-25\end{array}\right].$$

56. In $x\in(-2,\ 2)$

$\big||x|-1\big|$ is not differentiable at $x=-1,\ 0,\ 1$

$|\cos\pi x|$ is not differentiable

at $x=-\frac{3}{2},\ -\frac{1}{2},\ \frac{1}{2},\ \frac{3}{2}.$

57. $a_0C_0-a_1C_1+a_2C_2-a_3C_3+.....+a_{2012}C_{2012}$

$=a_0\left(C_0-C_1+C_2+.....+C_{2012}\right)$

$-d\left(C_1-2C_2+3C_3-4C_4+.....-{}^{2012}C_{2012}\right)$

$=a_0(0)-d\left(C_1-2C_2+3C_3-4C_4\right.$

$\left.+.....-{}^{2012}C_{2012}\right)$

$=a_0(0)-d(0)=0.$

58. $\left|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right|=\sum\left|\vec{a}\right|^2+2\sum\vec{a}\ .\ \vec{b}$

$=4+2\vec{d}\ .\ \left(\vec{a}+\vec{b}+\vec{c}\right)$($a$, b, c are mutually $\perp r$)

Let $\vec{d}=\lambda\vec{a}+\mu\vec{b}+v\vec{c}$

Then $\vec{d}\ .\ \vec{a}=\vec{d}\ .\ \vec{b}=\vec{d}\ .\ \vec{c}=\cos\theta$

$\therefore\ \lambda=\mu=v=\cos\theta$

Also

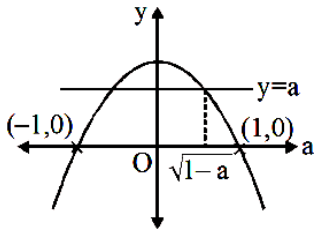
$\lambda^2+\mu^2+v^2=1=3\cos^2\theta$ or $\cos\theta=\pm\frac{1}{\sqrt{3}}$

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$\therefore \left|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right|^2 = 4 \pm 2 \cdot \frac{3}{\sqrt{3}} = 4 \pm 2\frac{3}{\sqrt{3}}.$

59.



$A(a) = 2\int_0^{\sqrt{1-a}} \left(\left(1-x^2\right)-a\right)dx = \frac{4}{3}(1-a)^{3/2}$

$\therefore A(0) = \frac{4}{3}$ and $A\left(\frac{1}{2}\right) = \frac{4}{3}\left(\frac{1}{2}\right)^{\frac{3}{2}} \Rightarrow \frac{A(0)}{A\left(\frac{1}{2}\right)} = 2\sqrt{2}.$

60. $(a-1)x + 0y + z = \alpha$ .....(1)

$x + (b-1)y + 0z = \beta$ .....(2)

$0x + y + (c-1)z = \gamma$ .....(3)

For no unique solution $D = 0$

$$\begin{vmatrix} (a-1) & 0 & 1 \\ 1 & (b-1) & 0 \\ 0 & 1 & (c-1) \end{vmatrix} = 0$$

$(a-1)(b-1)(c-1)+1 = 0$

$\therefore a = 2;\ b = 2;\ c = 0$ Hence, $|a+b+c| = 4.$

61. $\because$ $\arg(z) = \pi$ if $z = x + iy$

$\therefore$ $y = 0$ and $x < 0$

$\therefore$ $|z+3-i| = 1 \Rightarrow |x+3-i| = 1 \Rightarrow (x+3)^2 + 1 = 1 \Rightarrow x = -3$

$\therefore$ $z = -3 + i \cdot 0 \Rightarrow |z| = 3.$

62. $A^T = A$

$B^T = -B$

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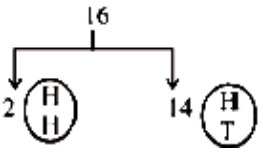
$$A-B=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \qquad \ldots(1)$$

$$A^T-B^T=\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$$

$$A+B=\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix} \qquad \ldots(2)$$

on adding eq. (1) and (2)

$$2A=\begin{bmatrix}2 & 5\\ 5 & 8\end{bmatrix} \qquad A=\begin{bmatrix}1 & 5/2\\ 5/2 & 4\end{bmatrix} \qquad |A|=\frac{4-25}{4}=\frac{-9}{4}.$$

63. 

$$\frac{2}{16}\times 1+\frac{14}{16}\times\frac{1}{2}=\frac{9}{16}.$$

64. $g\left(f(x)\right)=(\sin x+\cos x)^2-1=\sin 2x$

$$2x\in\left[-\frac{\pi}{2},\ \frac{\pi}{2}\right]$$

$$x\in\left[-\frac{\pi}{4},\ \frac{\pi}{4}\right].$$

65. $$I=\int\frac{\sec^2 x\,dx}{(4\tan x-5)^2}=\frac{1}{4}\int\frac{d(4\tan x-5)}{(4\tan x-5)^2}=\frac{-1}{4(4\tan x-5)}+C.$$

66. $$\frac{dy}{dx}=xy^2+x-y^2\sin x-\sin x \quad\Rightarrow\quad \frac{dy}{dx}=(x-\sin x)(y^2+1)$$

Hence, $\int\frac{dy}{y^2+1}=\int(x-\sin x)dx \Rightarrow \tan^{-1}y=\frac{x^2}{2}+\cos x+C$

So, $y=\tan\left(\frac{x^2}{2}+\cos x-1\right)$.

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67. Use $T = S_1$.

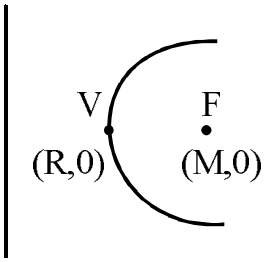
68. $V \rightarrow vertex$

$F \rightarrow focus$

$VF = M - R$

So length of latus rectum $= 4(M - R)$

$\Rightarrow$ Half of length of latus rectum $= 2(M - R)$



69. Let the other end of diameter is (h,k), then equation of circle is

$(x-h)(x-p)+(y-k)(y-q)=0$

Put $y = 0$, since $x-$ axis touches the circle

$\Rightarrow x^2-(h+p)x+(hp+ka)=0$

$\Rightarrow (h+p)^2 = 4(hp+kq)(D=0)$

$\Rightarrow (x-p)^2 = 4qy$

70. $l = \cos\theta, n = \cos\theta, m = \cos\beta$

$\cos^2\theta + \cos^2\theta + \cos^2\beta = 1$

$\Rightarrow 2\cos^2\theta = \sin^2\beta = 3 - 3\cos^2\theta$

$\Rightarrow 2\cos^2\theta = 3(1-\cos^2\theta) = 3 - 3\cos^2\theta$

$\Rightarrow 5\cos^2\theta = 3$

$\Rightarrow \cos^2\theta = \frac{3}{5} \Rightarrow \sec^2\theta = \frac{5}{3}$

71.

$\lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^+} f(x) \Rightarrow a+b=2$

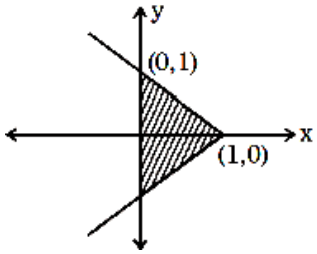
$\therefore \lim\limits_{x\to 3^-} f(x) = \lim\limits_{x\to 3^+} f(x) \Rightarrow 3a+b = 6\tan\frac{3\pi}{12}$

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$\therefore a=2,\ b=0.$

72.



$$\text{area} = 2\times\left(\frac{1}{2}\times1\times1\right)=1=k.$$

73. $2b=a+c \quad \ldots(1)$

$4a+2b+c=0 \quad \ldots(2)$

Let other roots is $\beta \quad 2\beta=\frac{c}{a}\Rightarrow\beta=\frac{c}{2a}$

From (1) and (2)

$4a+a+c+c=0\Rightarrow 5a=-2c$

$\frac{c}{a}=\frac{-5}{2}$ So, $\beta=\frac{-5}{4}=-1.25.$

74. $I=\int_2^4\frac{\log x^2}{\log x^2+\log(6-x)^2}dx \quad \ldots(1)$

By KING property $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)dx$

$I=\int_2^4\frac{\log(6-x)^2}{\log(6-x)^2+\log x^2}dx \quad \ldots(2)$

Adding (1) ((2)

$2I=\int_2^4 1\,dx=(x)_2^4=4-2=2 \quad \therefore\ I=1.$

75. $\vec{b}\,.\,\vec{c}=10\Rightarrow 5|\vec{c}|\cos\frac{\pi}{3}=10\Rightarrow|\vec{c}|=4$

$\left|\vec{a}\times\left(\vec{b}\times\vec{c}\right)\right|=\left|\vec{a}\right|\ \left|\vec{b}\times\vec{c}\right|\ =\sqrt{3}\,.\,5\,.\,4\,.\sin\frac{\pi}{3}=30.$

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